

The Kaiser Permanente/USC Patient Consultation Study: Change in quality of life

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Abstract: The impacts of two models of pharmacist consultation on patient function and health-related quality of life (HRQOL) relative to a control model were studied.

Patients in the random-assignment study and the area-wide study of the Kaiser Permanente/USC Patient Consultation Study were surveyed three times over a two-year period. The patients were receiving pharmaceuti-

cal services under the Kaiser Permanente model of consultation (KP model), a state model of consultation, or a control model and were stratified according to prescription drug use. A global visual-analogue scale and Short Form-36 were used to assess HRQOL.

In the areawide study, only 2 of 42 comparisons of the effects of the KP or state model on HRQOL were significant;

both were associated with the state model and involved only small increases. In the random-assignment study, 3 of 21 estimated effects of the KP model on HRQOL were positive and significant; 2 of these were significantly different between the KP model and the state model.

Although the KP and state models of consultation were associated with some changes in HRQOL, the overall influ-

ence was not consistent and not clinically important.

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Including measures of health-related quality of life (HRQOL) in outcome studies is important in order to ensure that the value of a medical intervention from the patient's perspective is considered.¹ Studies evaluating the influence of pharmacists' interventions on HRQOL have been limited, and findings have been mixed. Data from the Medical Outcomes Study (MOS) suggested that patients using insulin and depressed patients who adhered to recommendations about their medications were actually worse off than patients who did not comply.² More recently, a study of an intervention by independent community pharmacists showed significant improvement in several domains of quality of life in patients with hyperlipidemia.³ In contrast, an

evaluation of a model for providing pharmaceutical care to patients with type 2 diabetes showed no influence on quality of life.⁴ Clearly, more studies are needed.

The Kaiser Permanente/USC Patient Consultation Study was designed to investigate the impact of pharmacist consultations on a broad range of outcome measures, including HRQOL. Although there is consensus that HRQOL outcomes should be measured, there is no consensus on how best to accomplish this task. HRQOL is defined as "the value assigned to duration of life as modified by the impairments, functional states, perceptions, and social opportunities that are influenced by disease, injury, treatment, or policy."⁵ We used this definition in determining which HRQOL measures to

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select. We used the MOS Short Form-36 (SF-36) augmented with a visual-analogue scale for rating overall health status.

This article addresses the HRQOL outcomes of two models of pharmacist consultation: the Kaiser Permanente model (KP model) and the state model relative to a control model. The KP model provided focused, comprehensive pharmacist consultations for high-risk ambulatory care patients.⁶ The state model provided state-mandated pharmacist consultations for all new or changed prescriptions. Under the control model, counseling was provided at the pharmacist's discretion or the patient's request.

Methods

Study design. The design of the Kaiser Permanente/USC Patient Consultation Study has been described previously.^{7,8} HRQOL data were collected by mail survey from patients in a random-assignment study and an areawide study. In the random-assignment study, all patients who agreed to have their new or changed prescriptions filled at a single pharmacy were mailed surveys at baseline (January to March 30, 1993), after one year (March 1994), and at the completion of the study (February 1995). In the areawide study, a stratified random sample of 1500 patients in each of six medical service areas was mailed surveys during the same time frame. Patients were stratified according to their prescription drug use during calendar year 1992 into four groups: use of specific targeted medications, concurrent use of five or more long-term medications for chronic conditions, use of five or more medications including at least one targeted medication ("polypharmacy and targeted medications"), and use of fewer than five medications and no targeted medications ("low-risk").⁷

Over the two-year study period, many of the patients initially taking five or more long-term medications but no targeted medications had at least one targeted medication added to their regimen, making the group too small for data analysis. We thus looked at three of the original groups—low-risk, targeted medication, and polypharmacy and targeted medication—and added two additional high-risk groups: high-risk-drug and failing-health. The high-risk-drug group included patients who filled a prescription for a nonsteroidal anti-inflammatory drug or a drug for anxiety, chronic obstructive pulmonary disease, diabetes, or a seizure disorder. The failing-health group consisted of patients who reported their health status at baseline as "worse" or "much worse" than in the previous year.

The sample for this analysis included participants who had at least one new prescription filled during the two-year study period, completed the final survey or died before the final survey, and had relatively complete survey data for all the independent variables used in the statistical analyses.⁸

Instruments. A global visual-analogue scale and SF-36 were used to assess HRQOL. The global visual-analogue scale consisted of a single question: "How would you rate your overall health?"; the patient was asked to place an X on a 100-mm line between 0 (worse than death) and 100 (excellent). The score was then read in millimeters (range, 0–100).

SF-36 considers the interrelationships among disease, personal functioning, psychological distress or well-being, general health perceptions, and social role functioning.⁹ It is a 36-item self-administered survey that measures eight health concepts: physical functioning, social functioning, role limitations due to physical health, role limitations due to mental health, general health, emotional well-being, vitality, and bodily pain.¹⁰ The health concepts described in SF-36 range in score from 0 to 100, with higher scores indicating higher levels of function or better health. Because of wide variation in the scale values, it is recommended that normalized *t* scores (*z* scores multiplied by 10 with 50 added) supplement the 0–100 scores. The normalized scores generate a standard mean of 50 and a standard deviation of 10. These statistics provide a way to equate central tendency and variability across scales and to compare scores across populations.

The Medical Outcomes Institute developed two summary scores for SF-36, one for physical health and one for mental health.¹¹ The physical component score (PCS) consists of the scales for physical function, general health, bodily pain, and role limitations due to physical health. The mental health component score (MCS) consists of the scales for emotional well-being, role limitations due to mental health, social functioning, and vitality. Ware et al.¹¹ noted that 80–85% of the reliable variance in the eight SF-36 scales can be accounted for by these two components. We calculated both scale scores and summary scale scores for all patient risk groups included in this analysis.

SF-36 includes two additional measures of health status that reflect respondents' self-reported global assessment of their health status and their trend in health status (health transition). Health status was analyzed by using a single item: "In general, would you say your health is excellent, very good, good, fair, or poor?"¹² The health-transition item asks respondents to compare their current health status with their health status in the previous year. Responses range on a 5-point scale from "much better" to "much worse." The single-item health-status score was used to evaluate the influence of including patients who died.¹² The health-transition item was used to identify patients in the failing-health group.

Measurement of treatment outcomes. In a longitudinal study, changes in HRQOL outcomes over time can be unrelated to the interventions being evaluated.¹³ This is particularly true in a chronically ill population in which the natural course of the disease

can affect an individual's evaluation of overall health status. Furthermore, the magnitude of observed changes in health-status scores may depend on the patient's initial score.¹⁴ Two methods were used to control for these factors. First, a clinically relevant "effect size" was determined by comparing the magnitude of the observed change in health status with the variability of the health-status measure in stable patients, in this case the variability at baseline. Second, the change in each participant's score between baseline and year 2 was used to control for differences in baseline values and clinically relevant effect sizes calculated for the outcome measures.

Mortality was not expected to be a frequent outcome, because the sampling procedure excluded patients with known terminal conditions. For example, patients with prescriptions for drugs for AIDS or HIV infection and antineoplastic agents were excluded. As a result, there were relatively few deaths over the 23-month study period: 77 (1.3%) in the areawide study and 84 (1.8%) in the random-assignment study. Separate analyses were conducted for survivors only and for all participants. In the latter analysis, values were imputed for patients who died. Death was considered the worst state of health on the visual-analogue scale and was therefore assigned a value of zero.

In profile measurements of health status there is no specification of death as the worst state. We looked at the number of respondents at zero for each of the scales. In the total sample, 1.3% were at zero for the physical function scale, 1.1% for the bodily pain scale, 19% for the role limitations due to physical health scale, and 0.2% for the general health scale. On the mental health scales, 0.1% were at zero for the emotional well-being scale, 1.0% for the social functioning scale, 13.2% for the role limitations due to emotional health scale, and 1.3% for the vitality scale. Because there was no evidence of a floor effect in this sample, zero or worst outcome was assigned for all year 2 outcomes for patients who died within the study period.¹² We used specifications described by Diehr et al.¹² for participants who had died. The findings were consistent, so only the results with death scored as zero are reported.

Regression analysis. Regression analysis was used to estimate the effects of the KP and state models of pharmacist consultation on quality of life, controlling for patient characteristics and other factors. In the areawide study, the treatment variables (alternative models of pharmacist consultation) were specified as dichotomous variables, and patients assigned to the control model were used as the reference group. In the random-assignment study, the treatment variables were the number of new prescriptions filled at a KP-model pharmacy and the number of new prescriptions filled at a state-model pharmacy. New prescriptions filled at a control-model pharmacy were the reference

group. The independent variables included demographic characteristics, health status at baseline, health behaviors, chronic diseases, number of new medications, and health care use at baseline.⁸ Unless otherwise stated, all analyses included patients who died.

Analyses were performed by using SAS for Windows (release 6.12, SAS Institute, Cary, NC). The a priori level of significance was set at 0.05.

Results

Inclusion in the final sample for the HRQOL analysis required either a final survey or death of the patient and use of at least one prescription medication during the study period. This reduced the sample to 2892 (63% of the baseline sample) in the areawide study and 4498 (49%) in the random-assignment study. Missing data were dealt with in two ways. First, if 60% of the items in an SF-36 scale were completed, the score was calculated by using the average of the items completed.¹¹ If more than 60% of the items were missing, a scale was coded as missing.

Second, imputed values were created for missing scale scores for patients with only one missing scale value. Buck's method for imputing conditional means was used.¹⁵ The imputed values were based on statistical models that regressed each dimension of health status on the remaining seven dimensions and other patient characteristics. A random component was also added to the imputed value, which reflected the variation in the study population. Patients with two or more missing values were excluded from the analyses. This resulted in an additional 200 patients being excluded from the areawide-study sample and 363 being excluded from the random-assignment study sample. The final sample included 2692 patients in the areawide study and 4135 in the random-assignment study.

The baseline health status of the samples by risk group is presented in Table 1. The polypharmacy and targeted-medication group reported worse health status than the U.S. average; all the scale scores were below the 50-point national mean. The low-risk group was not different from the U.S. average. Table 2 compares the raw scores for the entire Kaiser Permanente/USC study sample with the raw scores for the U.S. population. Not surprisingly, the study sample, which was selected on the basis of higher than average use of prescription medications, reported poorer HRQOL across all domains.

The mean change in health status, as measured by the change in each of the HRQOL scales, is reported by consultation model for the areawide-study sample in Table 3. Data are not reported for the random-assignment study because this sample had high rates of crossover and a single consultation model could not be assigned.⁸ The mean change in scores over the 23-month study period ranged from -4.18 to 2.18. Most of the change was negative, which is to be expected in a

chronically ill population. However, the magnitude of the change was not clinically important. Ware¹⁶ reported that a change of ± 5 can be considered important,

Table 1.
Baseline Health Status by Risk Group^a

SF-36 Scale	Targeted Drug ^b	Polypharmacy and Targeted Drug ^c	Low Risk ^d
<i>Areawide Study</i>			
Bodily pain	46.5	41.7	48.8
General health	48.8	43.0	51.4
Physical function	47.9	40.9	50.1
Physical role	47.1	41.1	49.6
Mental health	49.4	48.0	50.3
Emotional role	49.3	47.3	51.1
Social functioning	48.3	44.0	50.0
Vitality	48.8	44.7	49.8
<i>Random-Assignment Study</i>			
Bodily pain	47.0	41.9	50.0
General health	49.1	42.8	52.3
Physical function	48.6	41.9	51.5
Physical role	48.3	42.9	50.9
Mental health	49.2	47.7	49.3
Emotional role	49.5	46.9	49.6
Social functioning	48.9	44.4	49.7
Vitality	49.0	45.4	50.2

^aReported as mean normalized *t* score on Short Form-36 (SF-36) scale.

^b*n* = 1528, areawide study; *n* = 2410, random-assignment study.

^c*n* = 718, areawide study; *n* = 858, random-assignment study.

^d*n* = 425, areawide study; *n* = 839, random-assignment study.

Table 2.
Comparison of Health Status of Kaiser Permanente/USC Study Sample with U.S. Population^a

SF-36 Scale	U.S. Population ^b	Total Kaiser Permanente Sample
Physical function	84.5	77.6
Physical role	81.2	70.7
Bodily pain	75.5	66.3
General health	72.2	68.1
Mental health	74.8	72.9
Social functioning	83.6	78.4
Emotional role	81.3	77.8
Vitality	61.1	57.4

^aReported as mean raw score on SF-36 scale.

^bSource: reference 11.

and there were no changes that met this criterion. In addition, the effect sizes ranged from 0.003 to 0.15, which would not be considered an important change in health status.¹⁷ When analysis of variance with a Bonferroni correction was used, there were no significant changes with any of the consultation models.

Results of the regression analyses are presented in Table 4. Only the analysis for the SF-36 summary scores are provided because the results for individual scales were consistent with the summary results. In the analysis of the results of the areawide study, only 2 of 42 comparisons of the effects of the KP or state consultation model versus the control model were significant. Both effects were seen with the state model of consultation and involved small increases (1.72 and 2.62 points) on the global visual-analogue scale.

In the larger random-assignment study sample, 3 of 21 estimated effects of the KP model of consultation were positive and significant. Estimated effects reflect changes for each new prescription filled at a KP-model pharmacy. Two effects were significantly different between the KP model and the state model: the change in the PCS and the change in the MCS in the total sample, including deaths. In the state model, most of the estimated effects were negative, indicating a reduction in HRQOL, although only 2 of 21 estimated effects (both involving the low-risk-group) were significant.

Discussion

The general lack of improvement in HRQOL was not surprising, because many factors influence quality of life. Point-of-service pharmacist consultation may not be intensive enough to alter patient drug-taking behavior sufficiently to produce a demonstrable, clinically important change in HRQOL. Furthermore, changes in HRQOL would be best measured by a series of closely spaced surveys triggered by an encounter with a pharmacist. In this study, the HRQOL survey was administered on a yearly basis, so any changes in HRQOL would have to have been sustained to be detected. Although a sustained change is desirable, it is not known whether better adherence to medication regimens and other

Table 3.
Change in Health Status of Areawide-Study Participants^a by Consultation Model

Short Form-36 Scale	Kaiser Permanente Model (<i>n</i> = 910)	State Model (<i>n</i> = 952)	Control Model (<i>n</i> = 808)
Physical component	-1.01 $\pm$ 9.35	-0.96 $\pm$ 8.59	-1.12 $\pm$ 9.31
Physical function	-2.61 $\pm$ 20.4	-3.59 $\pm$ 19.2	-3.31 $\pm$ 20.8
Bodily pain	0.30 $\pm$ 24.9	0.33 $\pm$ 22.9	-0.68 $\pm$ 25.6
Physical role	-4.17 $\pm$ 42.1	-1.07 $\pm$ 41.3	-4.18 $\pm$ 42.3
General health	-2.18 $\pm$ 18.0	-3.04 $\pm$ 16.5	-2.19 $\pm$ 16.9
Mental component	-0.04 $\pm$ 10.7	0.08 $\pm$ 9.85	-0.21 $\pm$ 9.98
Mental health	0.69 $\pm$ 17.1	0.59 $\pm$ 15.4	0.06 $\pm$ 16.4
Social functioning	-3.24 $\pm$ 28.7	-1.92 $\pm$ 27.2	-2.96 $\pm$ 28.8
Emotional role	-1.40 $\pm$ 40.1	-1.32 $\pm$ 39.3	-3.03 $\pm$ 39.0
Vitality	-0.39 $\pm$ 19.2	-0.12 $\pm$ 17.8	0.18 $\pm$ 18.5
Global visual-analogue scale	0.81 $\pm$ 18.4	2.18 $\pm$ 16.9	0.23 $\pm$ 18.0

^aReported as mean $\pm$ S.D. of the change in each participant's *t* score between baseline and year 2.

Table 4.
Estimated Effect of Consultation Model on Physical Health, Mental Health, and Health-Related Quality of Life (HRQOL)

Sample and Scale ^a	Areawide Study			Random-Assignment Study		
	<i>R</i> ²	Estimated Effect ^b		<i>R</i> ²	Estimated Effect ^b	
		Kaiser Permanente Model	State Model		Kaiser Permanente Model	State Model
All survivors ^c						
PCS	0.249	−0.06	−0.23	0.284	0.017	−0.027
MCS	0.307	0.28	0.45	0.310	0.068	0.021
Global HRQOL	0.098	1.42	1.72 ^d	0.074	0.004	−0.049
All participants ^e						
PCS	0.202	−0.17	−0.30	0.242	0.038 ^f	−0.079
MCS	0.202	−0.01	0.18	0.258	0.129 ^{d,f}	0.016
Global HRQOL	0.094	1.52	1.42	0.072	0.071	−0.053
Targeted-drug group ^g						
PCS	0.251	−0.01	−0.82	0.241	0.059	−0.010
MCS	0.257	0.81	1.04	0.275	0.123	−0.013
Global HRQOL	0.106	1.48	2.62 ^d	0.086	0.202	0.083
Polypharmacy and targeted-drug group ^h						
PCS	0.232	−1.63	−0.04	0.300	0.070	−0.018
MCS	0.252	−1.05	−0.53	0.263	0.162 ^d	0.064
Global HRQOL	0.177	0.12	−1.74	0.111	0.039	−0.055
Low-risk group ⁱ						
PCS	0.341	0.81	2.41	0.314	−0.268	−0.357 ^d
MCS	0.336	−0.28	−0.91	0.354	−0.216	−0.397 ^d
Global HRQOL	0.173	1.03	2.89	0.144	−0.183	−0.177
High-risk-drug group ^j						
PCS	0.216	−0.21	−0.41	0.255	0.071	−0.014
MCS	0.207	−0.08	0.23	0.253	0.118 ^d	0.042
Global HRQOL	0.104	1.22	1.06	0.081	0.068	−0.024
Failing-health group ^k						
PCS	0.369	−1.24	−2.09	0.389	0.038	−0.079
MCS	0.378	−1.61	−1.85	0.3120	0.168	0.011
Global HRQOL	0.229	2.51	−4.20	0.161	0.466	0.240

^aPCS = Short Form-36 (SF-36) physical component score. MCS = SF-36 mental health component score. Global HRQOL measured on visual-analogue scale for the question, How would you rate your overall health?

^bReported as β coefficient.

^c*n* = 2430, areawide study; *n* = 4058, random-assignment study.

^d*p* < 0.05 for indicated model versus control model.

^eIncluding patients who died during study. *n* = 2692, areawide study; *n* = 4134, random-assignment study.

^f*p* < 0.03, KP model versus state model.

^g*n* = 1384, areawide study; *n* = 2409, random-assignment study.

^h*n* = 642, areawide study; *n* = 857, random-assignment study.

ⁱ*n* = 395, areawide study; *n* = 838, random-assignment study.

^j*n* = 1752, areawide study; *n* = 2836, random-assignment study.

^k*n* = 349, areawide study; *n* = 537, random-assignment study.

modifications in a patient's drug regimen can substantially improve the health of a population with a variety of chronic diseases measured annually. Future research will need to look more closely at the influence of pharmacist consultations on patient behavior, avoidance of adverse drug reactions, and prevention or treatment of unwanted adverse effects and how these issues influence HRQOL.

Relatively few significant effects of pharmacist consultation were found, and those effects were different between the areawide and random-assignment studies. The analysis of the results of the areawide study could specify the treatment as a dichotomous variable because of the low rate of crossover between models of pharmacist consultation. The number of encounters with a pharmacist over the study period was accounted

for by including the total number of new prescriptions filled as an independent variable. The only significant findings were slight improvements in the state model for global HRQOL reported by patients using targeted medications and in the total sample excluding patients who died. The latter finding was not robust, because the differences disappeared when patients who died were included in the analysis. Furthermore, the magnitude of improvement did not pass the test for clinical meaningfulness.

The findings in the random-assignment study were varied and not always consistent with expectations. The KP model was associated with a small, significant positive effect on the mental-health component of HRQOL for the total sample (including deaths) and for two high-risk groups—patients taking five or more

long-term medications including at least one targeted medication and patients taking high-risk drugs. These risk groups were thought to be the two groups most likely to be influenced by the KP model of consultation; however, the consultation model was not predicted to influence psychological well-being. Once again, the small changes in HRQOL associated with pharmacist consultations did not meet the conditions for clinical meaningfulness.

Conclusion

Although the KP and state models of pharmacist consultation were associated with some changes in health-related quality of life, the overall influence was not consistent and not clinically important.

References

- Ware JE. Conceptualizing disease impact and treatment outcomes. *Cancer*. 1984; 53:2316-26.
- Hays RD, Kravitz RL, Mazel RM et al. The impact of patient adherence on health outcomes for patients with chronic disease in the medical outcomes study. *J Behav Med*. 1994; 17:347-60.
- Shibley MC, Pugh CB. Implementation of pharmaceutical care services for patients with hyperlipidemias by independent community pharmacy practitioners. *Ann Pharmacother*. 1997; 31:713-9.
- Jaber LA, Halapy H, Fernet M et al. Evaluation of a pharmaceutical care model on diabetes management. *Ann Pharmacother*. 1996; 30:238-43.
- Patrick DL, Erickson P. Health status and health policy: quality of life in health care evaluation and resource allocation. New York: Oxford Univ. Press; 1993.
- Johnson KA, Nye M, Hill-Besinque K et al. Measuring the impact of patient counseling in the outpatient pharmacy setting: development and implementation of the counseling models for the Kaiser Permanente/USC Patient Consultation Study. *Clin Ther*. 1995; 17:988-1002.
- McCombs JS, Cody M, Besinque K et al. Measuring the impact of patient counseling in the outpatient setting. The research design of the Kaiser Permanente/USC Patient Consultation Study. *Clin Ther*. 1995; 17:1188-206.
- McCombs JS, Liu G, Shi J et al. The Kaiser Permanente/USC Patient Consultation Study: use and cost of health care services. *Am J Health-Syst Pharm*. 1998; 55:2485-99.
- Ware JE. Conceptualizing disease impact and treatment outcomes. *Cancer*. 1984; 53:2316-26.
- Hays RD, Sherbourne CD, Mazel RM. The RAND 36-item health survey 1.0. *Economics*. 1993; 2:217-27.
- Ware JE, Kosinski M, Keller SD. SF-36 physical and mental health summary scales: a user's manual. Boston: The Health Institute, New England Medical Center; 1994.
- Diehr P, Patrick DL, Hedrick SC et al. Including deaths when measuring health status over time. *Med Care*. 1995; 33:AS164-72.
- Bouchet C, Guillemin F, Briancon S. Nonspecific effects in longitudinal studies: impact on quality of life measures. *J Clin Epidemiol*. 1996; 49:15-20.
- Stucki G, Daltroy L, Katz JN et al. Interpretation of change scores in ordinal clinical scales and health status measures: the whole may not equal the sum of the parts. *J Clin Epidemiol*. 1996; 49:711-7.
- Little RJA, Rubin DB. Statistical analysis with missing data. New York: John Wiley & Sons; 1987.
- Ware JE. Patient-based assessments of health outcomes. Paper presented at AHSR 13th Annual Meeting. Atlanta, GA; 1996 Jun 11.
- Kazis LE, Anderson JJ, Meenan RF. Effect sizes for interpreting changes in health status. *Med Care*. 1989; 27:S178-89.